Space Emanation Theory - Dark Energy - Rough draft 2

A fixed parameter calculation of the brightness trend

The same SET calculation brings the average supernova brightness closer to observation in every one of the fifteen redshift groups examined here. The groups contain 960 higher-redshift objects from Pantheon+ and 1,619 from DES-Dovekie, each compared with a nearby reference sample. We keep the source inputs and response prescription fixed across both catalogues. No SET coefficient is optimized against their brightness residuals.

The question is concrete: can the source law and the capacity relation, with a specified history of the collective field, calculate the brightness trend usually studied through dark-energy cosmology? We develop the expression, follow it to an observable, and show the complete grouped comparison. One source-clock prescription substantially improves the relative brightness calculation across the sample.

1 From the source law to a rate

Sources and field capacity

Space Emanation Theory proposes that matter establishes a burden in the field through which physical interactions occur. Gravitational burden, motion and the progression of local events participate in one finite capacity. A clock measures that progression through its physical ticks. We examine a homogeneous cosmological realization of these principles and calculate how its source and clock relations affect the brightness received from distant supernovae.

The source law and the local capacity relation are

$$Q^2 = 24\pi GMV, \quad S^2 + V_{\mathrm{rel}}^2 + c^2\beta^2 = c^2 . \quad (1)$$

Here M is the baryonic mass assigned to the source domain and V is its volume. The law specifies the positive source quantity Q, with units $\mathrm{m}^3\,\mathrm{s}^{-1}$, using reference time T. The gravitational burden S is expressed in speed units; S^2 is its contribution to the capacity budget. V_{rel} describes local motion relative to the field. The dimensionless factor β describes the progression of a physical clock relative to T, and c sets the capacity scale. Volume V and velocity V_{rel} are distinct quantities.

For a spherical domain, $V = 4\pi R^3/3$ and $M = \rho V$. The source law gives

$$Q = 4\pi\sqrt{2GMR^3}, \quad H_b \equiv \frac{Q}{3V} = \sqrt{\frac{8\pi G\rho}{3}} . \quad (2)$$

The second equality is direct algebra. To use this rate for physical evolution, we adopt $dV_m/dT = Q_m$ for a material domain containing a fixed collection of baryonic matter. The subscript m distinguishes this domain from the advancing collective region introduced in Section 2. Its volume scales as $V_m \propto a^3$, where a is the common scale factor and $a = 1$ at reception. Conserved material mass then gives $\rho = \rho_0 a^{-3}$.

$$H_T \equiv \frac{1}{a}\frac{da}{dT} = \frac{Q_m}{3V_m} = H_b(a) = H_{b0}a^{-3/2} . \quad (3)$$

The factor three comes from differentiating a volume proportional to a^3. At any given epoch, $Q/(3V)$ has the same value for domains with the same uniform density. A fixed material mass does not require a constant Q_m: the source law also contains the evolving volume. Present-day values carry the subscript zero.

The same process read by a physical clock

The time T is the reference used to express the source law. The variable t records the reading of the comoving physical clock used in this cosmological calculation. SET interprets its ticks as events realized within the available field capacity. A smaller clock factor means fewer ticks per reference second. The relation between these two time readings is part of the physical description being tested here.

A comoving clock follows the local homogeneous matter and has $V_{\mathrm{rel}} = 0$ in this construction. The capacity relation therefore gives $\beta = \alpha$ and

$$u \equiv \frac{S^2}{c^2}, \quad \alpha = \sqrt{1-u}, \quad dt = \alpha\, dT. \quad (4)$$

For $0 \le u < 1$, the clock factor is positive. The same source-driven volume increment can be expressed using either time reading:

$$dV_m = Q_m\, dT = Q_{\mathrm{normalized},m}\, dt. \quad (5)$$

Dividing by the physical clock interval gives the clock-normalized source rate,

$$Q_{\mathrm{normalized},m} = Q_m \frac{dT}{dt} = \frac{Q_m}{\alpha}. \quad (6)$$

The source law is expressed per reference time. A clock within the burdened field registers a smaller interval over the same process, so expressing that process per second of this clock gives a larger numerical rate. The amount assigned to a given physical interval is unchanged. For example, if $\alpha = 1/2$ throughout an interval, one reference second corresponds to half a clock second and the clock-normalized rate is $2Q_m$. The factor follows from the clock conversion; it is not an extra source contribution.

We use this reading to calculate the evolution measured by the comoving clock:

$$H \equiv \frac{1}{a}\frac{da}{dt} = \frac{1}{3V_m}\frac{dV_m}{dt} = \frac{Q_{\mathrm{normalized},m}}{3V_m} = \frac{H_b}{\alpha}. \quad (7)$$

For any selected domain in this homogeneous prescription, we write $Q_{\mathrm{normalized}} = Q/\alpha$; the domain label can be omitted when it is clear. As the collective burden changes, α changes, and the source rate is read differently by physical clocks. Section 2 calculates this history. Section 3 uses it in the distance and brightness calculation. The inverse clock factor is already included in $H(z)$, so it is not applied again to the final brightness.

2 The collective burden history

To calculate α through time, we specify the collective source region that supplies the burden. This test uses uniform density inside a moving spherical front of radius R_h, with $M_h = 4\pi\rho R_h^3/3$. The radius identifies the domain included in this burden readout. Its kinematic evolution, expressed in the same clock time t, is

$$\frac{dR_h}{dt} = HR_h + c\,. \quad (8)$$

The first term carries the front with the evolving scale; the second advances it locally at c. We adopt this evolution for the collective boundary. Its enclosed mass can change as it advances, whereas the material domain used to define the source-volume rate has fixed mass.

For the burden readout we retain SET's isotropic static source kernel: a mass element dm at distance r contributes $dS^2 = 2G\,dm/r$. At the centre of a uniform sphere, a shell of thickness dr contains $dm = 4\pi\rho r^2 dr$. Adding these shell contributions gives

$$S_c^2 = 2G\int_0^{R_h} \frac{\rho}{r}\,4\pi r^2\,dr = 4\pi G\rho R_h^2 = \frac{3GM_h}{R_h}\,. \quad (9)$$

This is the central burden of the collective sphere. The spherical exterior expression $S^2 = 2GM/r$ describes a different evaluation location; it is not substituted for this interior sum. Since $H_b^2 = 8\pi G\rho/3$, the central result becomes

$$u = \xi\frac{H_b^2 R_h^2}{c^2}, \quad \xi = \frac{3}{2}\,. \quad (10)$$

The factor $3/2$ is the central uniform-source factor, fixed by this readout rather than adjusted numerically. We apply the static uniform-source kernel to the evolving domain as the burden prescription for this calculation. Together with the front evolution and source-volume law, it specifies the physical realization being tested.

Differentiate the burden expression with respect to $\ln a$. The density contributes -3, while the advancing radius contributes $2\,d\ln R_h/d\ln a$. Using the front and clock relations gives

$$\frac{du}{d\ln a} = u\left[-3 + 2\left(1 + \frac{c}{HR_h}\right)\right] = -u + 2\sqrt{\xi u(1-u)}\,. \quad (11)$$

The substitution $y = \sqrt{u/(1-u)}$ makes the numerical evolution especially simple:

$$\frac{dy}{d\ln a} = \frac{1+y^2}{2}\left(2\sqrt{\xi} - y\right), \quad H = H_{b0}a^{-3/2}\sqrt{1+y^2}\,. \quad (12)$$

This is the response equation used for the brightness calculation. The source law supplies the density-dependent rate; capacity supplies the clock conversion; the stated uniform-domain and front prescriptions supply the burden history. Once those prescriptions are retained, the differential equation follows without an adjustable dimming exponent.

3 From the history to observed brightness

For the photon calculation we retain the homogeneous, spatially flat propagation coupling, with photon number conserved. It relates emission and reception by $1 + z = a_o/a_e$, stretches clock intervals by $dt_o/dt_e = 1 + z$, and reduces each photon's received energy by the same factor. With $a_o = 1$, the luminosity distance and bolometric flux are

$$D_L(z) = (1 + z)c \int_0^z \frac{dz'}{H(z')}, \quad F_o = \frac{L_e}{4\pi D_L^2}. \quad (13)$$

In catalogue calculations, the integral uses z_{HD} and the prefactor uses $1 + z_{\mathsf{HEL}}$, following the published redshift convention. The released brightnesses already include light-curve standardization and observational corrections [1–4]. The test compares these standardized brightnesses across different supernovae, rather than the fading of one explosion over several days.

Define $E(z) = H(z)/H_0$ and a dimensionless luminosity distance $d_L = H_0 D_L/c$. The tested history gives

$$E(z) = (1 + z)^{3/2} \sqrt{\frac{1 + y(z)^2}{1 + y_0^2}}, \quad d_L = (1 + z_{\mathsf{HEL}}) \int_0^{z_{\mathsf{HD}}} \frac{dz'}{E(z')}. \quad (14)$$

The initial source-volume calculation has a constant clock factor, so its normalized history is $E_0(z) = (1 + z)^{3/2}$. Its distance is

$$d_{L,0} = 2(1 + z_{\mathsf{HEL}})\left[1 - (1 + z_{\mathsf{HD}})^{-1/2}\right]. \quad (15)$$

This is the matter-only, spatially flat baseline with a constant clock factor. Its normalized distance shape is also the classical dust cosmology shape. The comparison tests how an evolving SET capacity factor changes this initial prediction while the source inputs are held fixed.

For each target group, let $\mathscr{L}_i[x]$ mean its weighted mean of x minus the weighted mean in the fixed nearby reference group, $0.025 < z_{\mathsf{HD}} < 0.10$. Its weights sum to zero. Therefore an unknown constant luminosity or distance zero point cancels. If m denotes the released standardized magnitude or distance modulus, the plotted quantities are

$$B_{\mathsf{obs},i} = 10^{-0.4\mathscr{L}_i[m - 5\log_{10} d_{L,0}]}, \quad (16)$$

$$B_{\mathsf{SET},i} = 10^{-0.4\mathscr{L}_i[5\log_{10}(d_L/d_{L,0})]}. \quad (17)$$

The initial prediction is $B_0 = 1$. A value below one means dimmer than the low-redshift-anchored initial curve. The predicted contrast uses the catalogue redshifts and the fixed source calculation; observed brightness does not enter its numerator. Every object is evaluated at its own redshift before averaging. These are geometric-mean relative fluxes, because averaging is performed in magnitudes.

Why this brightness trend motivated dark energy

Type Ia supernovae allow us to compare received brightness with the brightness predicted at a measured redshift. Their light curves are standardized to account for known luminosity differences, and nearby supernovae establish a common reference. The question is then whether the predicted distance–redshift relation accounts for the brightness of the more distant sample.

The original supernova studies found distant events to be systematically fainter than expected in the matter-only cosmologies they examined. With standardized luminosity L, the flux relation $F = L/(4\pi D_L^2)$ assigns a larger luminosity distance to a fainter source. Fits to the full distance–redshift relation favored histories with recent accelerated expansion. In GR cosmology, a positive cosmological constant could produce that behavior. The studies compared competing expansion histories and assessed observational systematics, making acceleration a quantitative inference from the brightness data [7,8].

The modern comparison below uses a specified matter-only, spatially flat baseline. It illustrates the brightness departure with the present catalogues; it is not a reconstruction of the original discovery analysis. For the Pantheon+ group near $z = 0.731$, the baseline predicts relative brightness $B_0 = 1$, while the observed value is $B_{\mathrm{obs}} = 0.700656$. The observation is therefore 29.93% dimmer than this baseline after the nearby-reference normalization.

The SET history predicts $B_{\mathrm{SET}} = 0.706666$, or 29.33% dimmer than the same baseline. The corresponding effective relative distance factors are

$$B_{\mathrm{obs}}^{-1/2} = 1.1947, \quad B_{\mathrm{SET}}^{-1/2} = 1.1896\,. \quad (18)$$

These are group contrasts relative to the nearby reference, not absolute distances to a single supernova. The baseline predicts too much brightness, and the SET calculation reduces it by almost the amount required. Section 4 shows the source inputs and numerical steps. Its 42.72% and 0.86% discrepancies use observed brightness as the denominator; the dimming percentages above instead use the baseline.

The brightness departure that motivated dark-energy cosmology is also the observational target of this SET calculation. Here, the evolving collective burden and its clock conversion change the predicted distance–redshift relation without inserting an additional dark-energy density. The same prescription moves the predictions toward observation in all fifteen redshift groups. Across the full grouped comparison, the median absolute discrepancy falls from 38.53% to 2.09% in Pantheon+ and from 36.39% to 2.96% in DES.

4 A worked calculation

Measured inputs and the source calculation

Use the Pantheon+ group $0.65 \le z < 0.80$: 63 supernovae with weighted mean redshift 0.73085168, compared with the fixed 311-object nearby reference. This is the closest central group match in the run; the full sample follows in Figure 1 and Table 1. The present-day source inputs are

$$M_0 = 1.5005279083 \times 10^{53}\ \mathrm{kg}, \quad R_0 = 4.3992396698 \times 10^{26}\ \mathrm{m}. \quad (19)$$

The radius is 46.5 billion light-years. These frozen benchmark inputs use a cosmology-conditioned radius and imply

$$V_0 = 3.5663295987 \times 10^{80}\ \mathrm{m^3}, \quad Q_0 = 5.1893886387 \times 10^{62}\ \mathrm{m^3\,s^{-1}}, \quad (20)$$

$$H_{b0} = 14.96662955\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \quad u_0 = \frac{3GM_0}{R_0c^2} = 0.7598922704. \quad (21)$$

Thus $\alpha_0 = 0.4900078873$, $y_0 = 1.7789876897$, and $H_0 = 30.54365029\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. The initial condition is fixed before calculating the brightness contrast.

Expected matter only baseline

Evaluate the initial distance expression at every target and reference redshift. After the common reference normalization, this baseline has $\Delta m_0 = 0$ and expected relative brightness $B_0 = 1$. The value one is a defined comparison scale, not an absolute flux measurement.

Expected SET calculation

Integrate the burden history from the fixed initial condition, calculate the distances, and average the magnitude differences using the same weights. This gives

$$\Delta m_{\mathrm{SET}} = \mathscr{L}[5\log_{10}(d_L/d_{L,0})] = 0.37696377\ \mathrm{mag}, \quad (22)$$

$$B_{\mathrm{SET}} = 10^{-0.4(0.37696377)} = 0.70666648. \quad (23)$$

Observed brightness and comparison

The catalogue gives $\Delta m_{\mathrm{obs}} = 0.38623777 \pm 0.03403476\ \mathrm{mag}$, equivalent to $B_{\mathrm{obs}} = 0.70065608$ with marginal 1σ interval 0.67903317–0.72296754. Define the signed discrepancy as $100(B_{\mathrm{pred}}/B_{\mathrm{obs}} - 1)$.

Calculation	Relative brightness	Discrepancy
Initial matter-only baseline	1.00000000	+42.72%
SET source-clock calculation	0.70666648	+0.86%
Observed group	0.70065608	—

The remaining difference is 0.00927399 mag, or 0.27 marginal standard deviations. The SET calculation reduces a 42.72% central discrepancy to 0.86%, placing the prediction comfortably inside the observed 1σ interval.

5 The same calculation across both catalogues

Figure 1 shows the unchanged calculation for every fixed group. No group is removed for disagreeing. The orange line connects the group predictions; it is not a curve fitted through the observations.

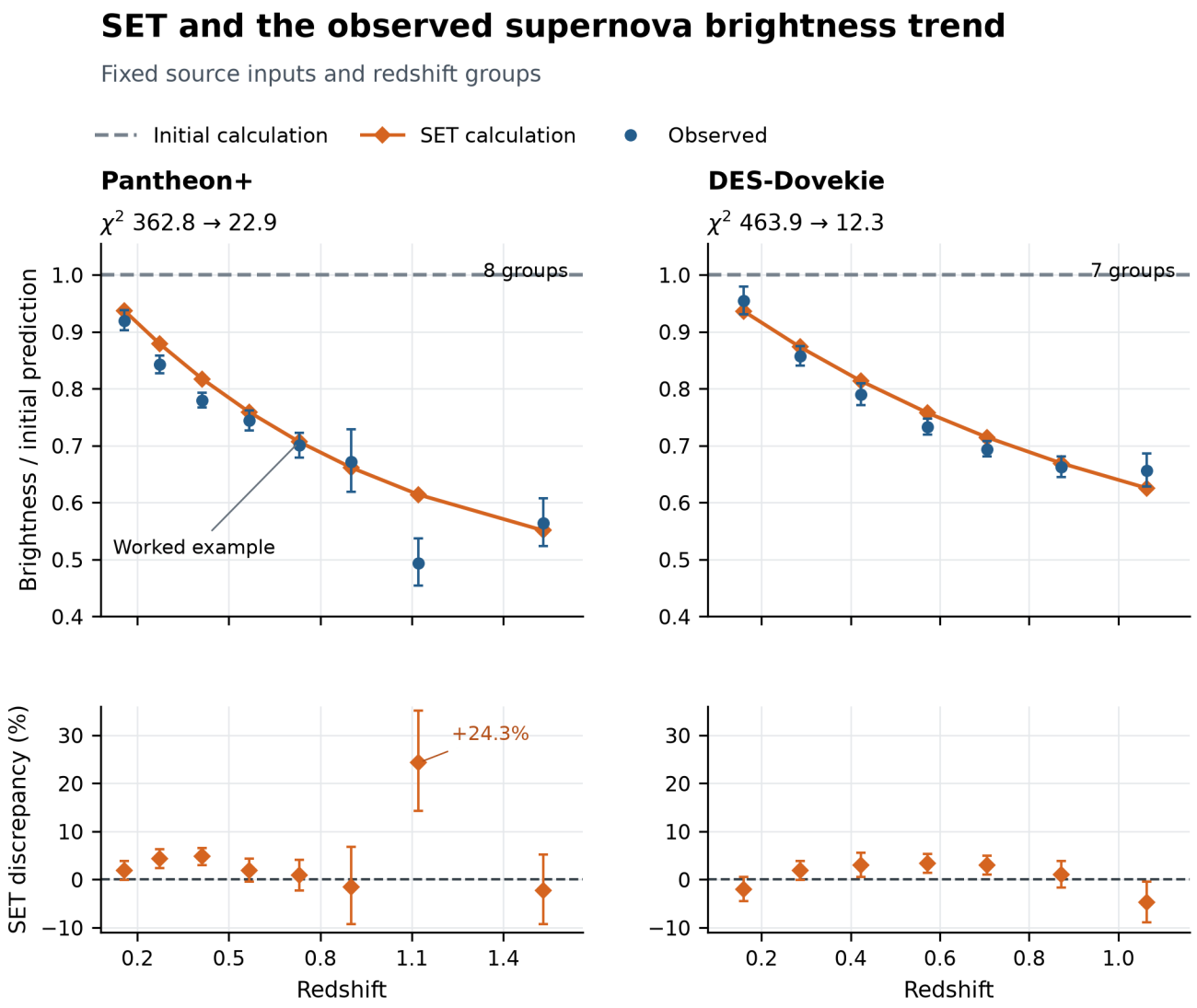


Figure 1. Relative supernova brightness and remaining SET discrepancy. Blue bars are marginal 1σ observational intervals. The initial source-volume prediction is one after reference normalization. The lower panels retain the sign of the remaining discrepancy. Formal comparisons use the full correlated magnitude covariance, including the common nearby reference. Source inputs are held fixed.

Comparison	Pantheon+	DES-Dovekie
Initial χ^2	362.75	463.91
SET χ^2	22.87	12.34
Median absolute error before	38.53%	36.39%
Median absolute error after	2.09%	2.96%

All fifteen central discrepancies decrease in magnitude. Eight group predictions lie within their marginal 1σ intervals: five of eight in Pantheon+ and three of seven in DES. These counts describe grouped means, not the fraction of individual supernovae explained. The two catalogue statistics must not be added as independent evidence.

6 Complete grouped results

Table 1 retains all plotted values. N is the number of distinct catalogue object identifiers in the target group; $\bar{z}$ is its weighted mean redshift. The observed interval is marginal 1σ. The initial prediction is one in every row. Before and after are signed percentage discrepancies relative to the observed central brightness.

Data	$\bar{z}$	N	Observed and 1σ interval	SET	Before	After
P	0.1575	207	0.919799 [0.902394, 0.937540]	0.936955	+8.72%	+1.87%
P	0.2729	380	0.842757 [0.827165, 0.858642]	0.879009	+18.66%	+4.30%
P	0.4131	163	0.779774 [0.766711, 0.793059]	0.817031	+28.24%	+4.78%
P	0.5675	117	0.744370 [0.726833, 0.762331]	0.758733	+34.34%	+1.93%
P	0.7309	63	0.700656 [0.679033, 0.722968]	0.706666	+42.72%	+0.86%
P	0.9018	5	0.671954 [0.619381, 0.728991]	0.661347	+48.82%	-1.58%
P	1.1217	9	0.494130 [0.454592, 0.537105]	0.614079	+102.38%	+24.27%
P	1.5302	16	0.563875 [0.523682, 0.607152]	0.551170	+77.34%	-2.25%
D	0.1608	44	0.954695 [0.930655, 0.979356]	0.935815	+4.75%	-1.98%
D	0.2863	309	0.857211 [0.840490, 0.874264]	0.873224	+16.66%	+1.87%
D	0.4225	404	0.789908 [0.771064, 0.809211]	0.813755	+26.60%	+3.02%
D	0.5726	467	0.733203 [0.719298, 0.747377]	0.757433	+36.39%	+3.30%
D	0.7059	265	0.694030 [0.680790, 0.707528]	0.714577	+44.09%	+2.96%
D	0.8721	113	0.662420 [0.644568, 0.680767]	0.669145	+50.96%	+1.02%
D	1.0639	17	0.656663 [0.628014, 0.686618]	0.625400	+52.29%	-4.76%

Table 1. Full results of the retained central uniform-burden calculation. P denotes Pantheon+ and D denotes DES-Dovekie. Values are rounded for presentation; calculations use the full stored precision.

The target intervals are [0.10, 0.20), [0.20, 0.35), [0.35, 0.50), [0.50, 0.65), [0.65, 0.80), [0.80, 1.00), [1.00, 1.30), and [1.30, 2.40). DES has no object in the final interval. The nearby references contain 311 Pantheon+ objects and 201 DES objects. Pantheon+ rows below $z = 0.025$ are not used in this fixed-reference plot.

Within each group and reference, weights use the published marginal inverse variance, with duplicate rows sharing object weight. The full statistical and systematic covariance is then propagated. This weighting was adopted in a documented amendment after an equal-object calculation exposed extremely uncertain DES entries; the redshift boundaries were unchanged. No weight or inclusion rule depends on a SET residual.

The sparse Pantheon+ group near $z = 1.122$ remains a visible exception to close central agreement: the discrepancy falls from +102.38% to +24.27%. Its nine objects remain in the table and figure.

7 The SET interpretation of the result

SET's physical proposal is that the baryonic sources participate in the same field capacity through which clocks, motion and light are realized. In this calculation, the source law fixes a rate per reference time. The collective burden determines how that reference rate is read by physical clocks. The moving front changes the burden history, and the history changes the integrated distance associated with a measured redshift. Photon energy, arrival rate and geometric spreading then determine the brightness.

The specific realization tested here includes a changing physical scale a. The brightness change comes from the coupled history of source-driven motion, clock progression and light propagation. No additional dark-energy density is inserted into these equations. A static delay along the same path would not by itself give this result: the operative quantity here is the evolving distance–redshift relation.

This connects to SET's broader motivation. The galaxy calculation investigates whether baryonic organization can alter the gravitational response through collective burden and effective area. Here the question concerns the history of a collective response and the light received through it. The source–capacity connection motivates both investigations, while the specific mathematical assumptions and observations differ.

The empirical value is the breadth of the improvement. A single fixed expression reduces the discrepancy in every group across two published compilations. The data were included by redshift and published quality information, rather than agreement with SET. The result concerns a trend supported by many observations, rather than one fortunate match. The median absolute central discrepancy falls from 38.53% to 2.09% in Pantheon+ and from 36.39% to 2.96% in DES.

The original research also checked exact catalogue overlap. Removing the 339 shared identifiers from both catalogues, while retaining the same response and redshift intervals, leaves a reduction in the correlated brightness statistic from 265.77 to 18.54 in Pantheon+ and from 187.09 to 8.82 in DES. The latter reference then contains only ten objects, and common calibration remains. This supports persistence of the improvement, but is not a blind independent replication.

This section establishes a result for relative brightness under the stated source, clock and propagation prescription. Its significance for SET is that the proposed connection has become calculable: the source law supplies a rate, the capacity relation changes its clock reading, and the resulting history produces the observed direction and much of the magnitude of the brightness departure. The same expression improves the full grouped comparison without a separate correction chosen for each group.

8 Data and reproducibility

The plotted measurements come from released standardized supernova products, not raw photon counts. Pantheon+ uses the published SN-only cut $z_{\mathrm{HD}} > 0.01$ before the fixed reference and group definitions are applied. DES uses the 1,820-row Dovekie release. The plotted target and reference groups contain 1,271 distinct Pantheon+ identifiers and all 1,820 DES identifiers, with overlap between catalogues.

For the formal statistic, stack the weighted contrasts into a matrix $\mathscr{L}$. If C is the released magnitude covariance and r the residual vector, the calculation is

$$C_{\mathrm{contrast}} = \mathscr{L} C \mathscr{L}^{\mathsf{T}}, \quad \chi^2 = r^{\mathsf{T}} C_{\mathrm{contrast}}^{-1} r . \quad (24)$$

The DES release supplies an inverse covariance, which is solved against to form the contrast covariance. It is not treated as the covariance itself. Error bars in Figure 1 are the marginal diagonal uncertainties, whereas this statistic includes correlations. The uncertainty bands are conditional on the frozen SET inputs; they are not model-parameter uncertainty bands.

A separate unbinned check retained the full released covariance and analytically profiled one constant luminosity intercept. Its initial and SET statistics were 2101.15 and 1426.45 for Pantheon+, and 2558.52 and 1647.64 for DES. This is a different check from the fixed-reference figure: it fits one nuisance zero point, but no SET parameter. It supports a broad improvement without establishing that every object improves.

The companion verification script recomputes all fifteen predictions directly from the fixed history and the catalogue redshifts, then checks them against the archived table. The forward predictions agree to better than 10^{-12} in relative brightness. The source history has also been cross-checked against an analytic integral of its evolution equation. The source records, plot and input hashes are retained with this section.

The research compared several physical prescriptions, including central and boundary burden readouts, before retaining this expression. Source parameters were held fixed, but this was exploratory development rather than a blind prediction. The reported χ^2 values describe these comparisons and are not corrected for the search across candidate histories.

Other calculated outputs define the scope of this realization: $H_0 = 30.54\ \mathrm{km\ s^{-1}\ Mpc^{-1}}$, compared with the megamaser measurement 73.9 ± 3.0 [5]; $\chi^2 = 48.70$ for six BAO geometry quantities [6]; and a physical domain ending at $z = 2.7244$. The relative brightness agreement reported here is therefore distinct from simultaneous agreement with those independent observables.

References

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